

Reducing New-Customer Churn: Executive Summary

Capstone Business Analysis — Customer Retention

Current Churn Rate	10.6%
Where Churn Happens	100% of churn occurs in a customer's first 11 months
Churn Rate, First Year	63.1%
Churn Rate, After Year 1	0.0%
Model Recall (catches at-risk customers)	100% on held-out test data
Highest-Risk Segment	21.8% churn vs. ~4.5% in the other two segments

The Problem

10.6% of customers churn. Every single churned customer left within their first 11 months — churn beyond that point is effectively zero. This is a first-year retention problem, not a general churn problem.

What Drives It

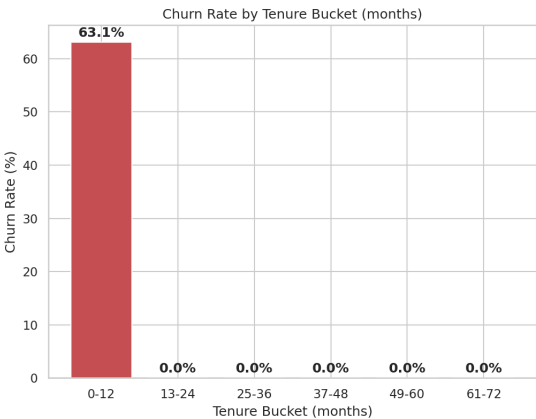
Statistically confirmed drivers (hypothesis-tested, not just correlated): **contract length** (month-to-month customers churn far more, $p < 0.001$) and **tenure itself** ($p < 0.001$). Monthly charges have a smaller but still significant effect. Payment method and senior-citizen status do not significantly affect churn.

What We Built

A predictive model flags at-risk customers with 100% recall (catches every churner in testing) at 65% precision. A simpler finding: a one-line rule (“flag anyone under 12 months tenure”) achieves nearly identical performance — the business can act immediately without waiting on a complex model. Customer segmentation also identified one segment (higher monthly charges, lower tenure) with 21.8% churn vs. ~4.5% elsewhere.

Recommended Actions

- **Immediate (Weeks 1-2):** Deploy the “Tenure < 12 months” flag in the CRM.
- **Short-term (Weeks 3-6):** Offer an incentive for at-risk customers to move onto an annual contract.
- **Medium-term (Weeks 5-10):** Build a dedicated onboarding track for the high-risk customer segment.
- **Ongoing:** Retrain the model quarterly; track whether the churn rate for the 0-12-month cohort falls toward parity with the rest of the base.



Full methodology, statistical tests, model details, and segmentation are in the Technical Report and `capstone_analysis.ipynb`.